

Geometry Theorem Template

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1 Prop. 4

1.1 Theorem Statement

Theorem 1 (Prop. 4: SAS Congruence). *If ABC and DEF are two triangles such that $\overline{AB} \cong \overline{DE}$, $\overline{AC} \cong \overline{DF}$, and $\angle BAC \cong \angle EDF$, then $\triangle ABC \cong \triangle DEF$.*

1.2 Diagram

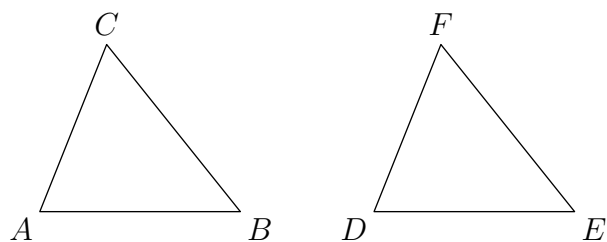


Figure 1: SAS Congruence

2 Prop. 5

2.1 Theorem Statement

Theorem 2 (Prop. 5: Isosceles Triangle Theorem Variant). *If $\overline{AB} \cong \overline{AC}$, then $\angle DBC \cong \angle ECB$.*

2.2 Diagram

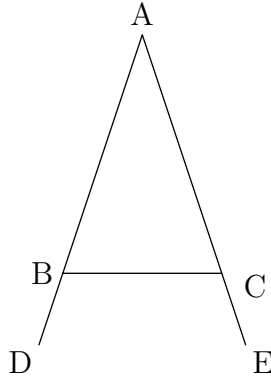


Figure 2: Variant of the Isosceles Triangle Theorem

2.3 Corollaries or Additional Notes

Corollary 1. *Euclid does not prove the version of the isosceles triangle theorem that is normally taught today.*

3 Prop. 6

3.1 Theorem Statement

Theorem 3 (Prop. 6: Converse of the Isosceles Triangle Theorem). *If $\triangle ABC$ is a triangle such that $\angle ABC \cong \angle ACB$, then $\overline{AB} \cong \overline{AC}$.*

3.2 Diagram

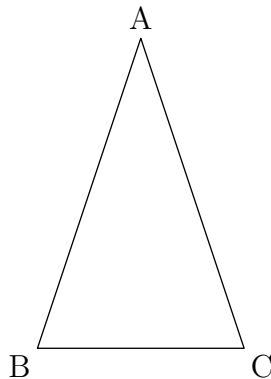


Figure 3: Converse of the Isosceles Triangle Theorem

3.3 Corollaries or Additional Notes

Corollary 2. *Any additional corollaries or notes can go here.*